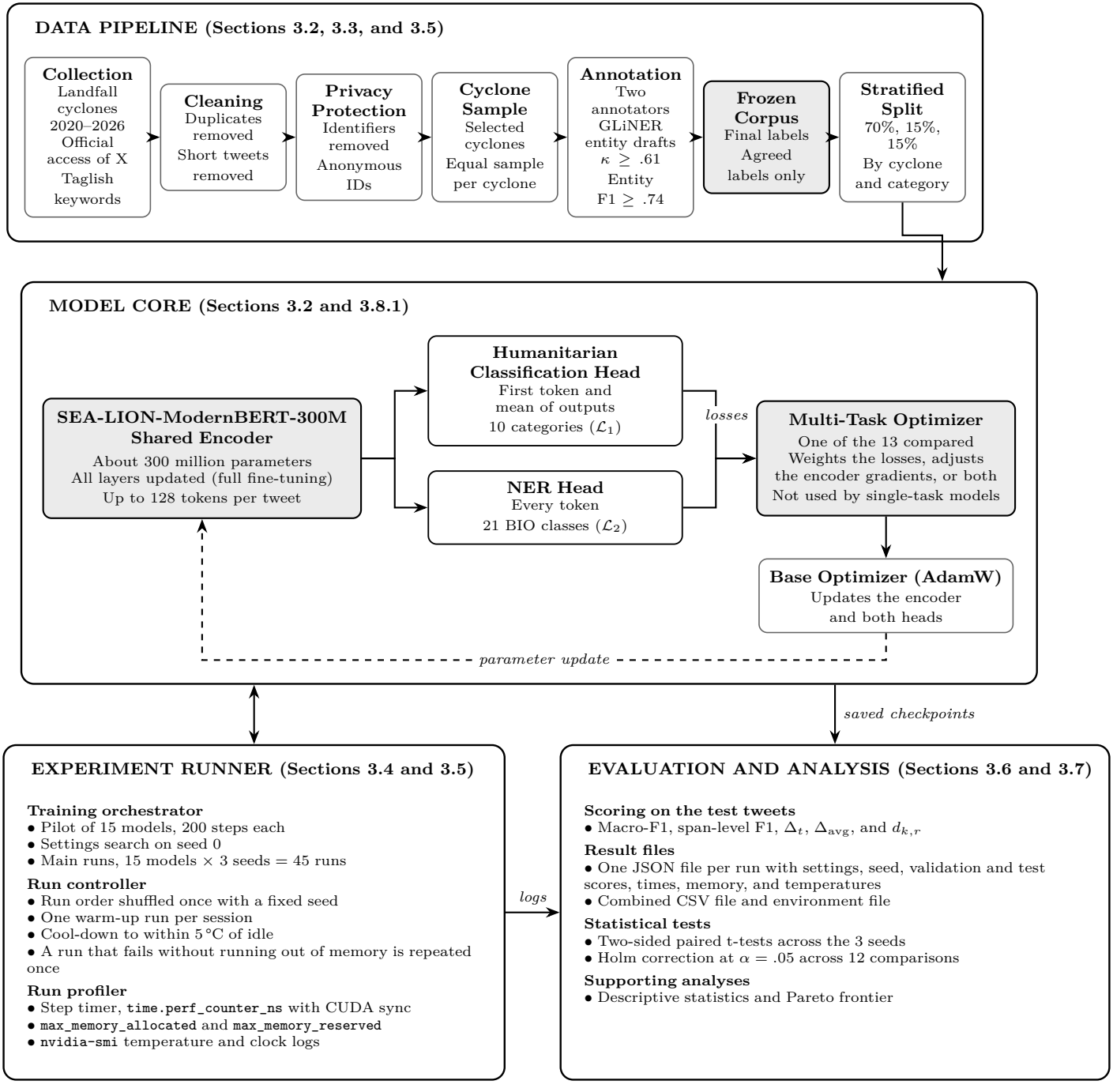
DATA PIPELINE (Sections 3.2, 3.3, and 3.5)
Collection
Landfall cyclones 2020–2026
Official access of X
Taglish keywords
Cleaning
Duplicates removed
Short tweets removed
Privacy Protection
Identifiers removed
Anonymous IDs
Cyclone Sample
Selected cyclones
Equal sample per cyclone
Annotation
Two annotators
GLiNER entity drafts
$\kappa \geq .61$
Entity F1 $\geq .74$
Frozen Corpus
Final labels
Agreed labels only
Stratified Split
70%, 15%, 15%
By cyclone and category
MODEL CORE (Sections 3.2 and 3.8.1)
SEA-LION-ModernBERT-300M Shared Encoder
About 300 million parameters
All layers updated (full fine-tuning)
Up to 128 tokens per tweet
Humanitarian Classification Head
First token and mean of outputs
10 categories ($\mathcal{L}_1$)
NER Head
Every token
21 BIO classes ($\mathcal{L}_2$)
losses
Multi-Task Optimizer
One of the 13 compared
Weights the losses, adjusts the encoder gradients, or both
Not used by single-task models
Base Optimizer (AdamW)
Updates the encoder and both heads
parameter update
saved checkpoints
EXPERIMENT RUNNER (Sections 3.4 and 3.5)
Training orchestrator
• Pilot of 15 models, 200 steps each
• Settings search on seed 0
• Main runs, 15 models × 3 seeds = 45 runs
Run controller
• Run order shuffled once with a fixed seed
• One warm-up run per session
• Cool-down to within 5 °C of idle
• A run that fails without running out of memory is repeated once
Run profiler
• Step timer, time.perf_counter_ns with CUDA sync
• max_memory_allocated and max_memory_reserved
• nvidia-smi temperature and clock logs
logs
EVALUATION AND ANALYSIS (Sections 3.6 and 3.7)
Scoring on the test tweets
• Macro-F1, span-level F1, Δ_t, Δ_{avg}, and $d_{k,r}$
Result files
• One JSON file per run with settings, seed, validation and test scores, times, memory, and temperatures
• Combined CSV file and environment file
Statistical tests
• Two-sided paired t-tests across the 3 seeds
• Holm correction at $\alpha = .05$ across 12 comparisons
Supporting analyses
• Descriptive statistics and Pareto frontier